Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Final Report

Group 9

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1. Abstract

The proliferation of misinformation on social media platforms and news websites poses significant risks to public trust and knowledge dissemination. Existing detection systems often focus on text alone, overlooking the visual elements that frequently accompany misleading content.

This project proposes a **Multimodal Misinformation Detector**, integrating Natural Language Processing (NLP) and Computer Vision (CV) techniques to identify misinformation across both textual and visual modalities. The system leverages evidence retrieval from external APIs, fine-tuned transformer models, and explainable AI techniques to deliver both classification and reasoning. The final implementation achieves an accuracy of **95% on both text verification** and **image authenticity detection**, and is deployed as an interactive **Hugging Face Space** with an intuitive Gradio-based interface..

2. Introduction

2.1 Problem Statement

Misinformation detection remains an open challenge, especially when content is multimodal—combining both text and images. Traditional NLP-based fact-checking systems do not generalize well to social media posts containing memes or manipulated images. Hence, this project addresses the need for a unified framework to detect misinformation from both textual and visual sources.

2.2 Motivation

The rise of multimodal misinformation can undermine media integrity and influence public opinion. Detecting and classifying such content requires models capable of understanding textual semantics, visual features, and contextual cues. This motivated the design of a hybrid pipeline combining transformer-based models and external retrieval APIs.

2.3 Objectives

- Develop a fact-verification model for textual claims using transformer architectures.
- Implement an image authenticity classifier using a fine-tuned CLIP model.
- Integrate both modalities into a unified inference pipeline with a clear, explainable output.
- Deploy the model on Hugging Face Spaces for public accessibility.

3. Literature Review

Research in misinformation detection has progressed along two main fronts — image deepfake detection and text-based fact verification. In visual forensics, large-scale datasets such as **FaceForensics++** (Rössler et al., 2019) and the **DeepFake Detection Challenge** (Dolhansky et al., 2020) enabled benchmarked evaluation using CNN-based detectors like XceptionNet, achieving AUCs near 0.90 but showing reduced generalization to unseen manipulations. Lightweight architectures such as **MesoNet** (Afchar et al., 2018) and GAN-fingerprint methods (Marra et al., 2019) improved efficiency, yet most relied solely on visual cues. Recent advances, notably **CLIP** (Radford et al., 2021), highlight how vision–language alignment can improve robustness against cross-domain fakes.

For textual misinformation, automated fact-checking typically follows a **Retrieve–Rerank–Verify** pipeline. The FEVER dataset (Thorne et al., 2018) established this paradigm, later extended by **FEVEROUS** (Aly et al., 2021) to include structured and multimodal evidence. Transformer models such as **BERT** and **DeBERTa** (Zhou et al., 2019) have become standard, yet retrieval recall and explainability remain major challenges (Kotonya & Toni, 2020; Guo et al., 2022). More recent retrieval-augmented generation (RAG) frameworks (Lewis et al., 2020) integrate evidence with large language models to improve interpretability, though factual grounding and evidence freshness are ongoing concerns.

This project builds upon these findings by proposing a unified **multimodal credibility detection pipeline** that combines CLIP-based image analysis, DeBERTa-based text verification, multi-source evidence retrieval (Google Fact Check, Tavily, Wikipedia), and LLM-generated explanations, thereby addressing the limitations of unimodal and non-explainable approaches.

4. Dataset and Methodology

4.1 Overview

The project integrates two complementary datasets — one for image-based fake detection and another for text-based claim verification. The combination enables training and evaluation of both visual and linguistic credibility models within a unified multimodal misinformation detection framework.

4.2 Image Dataset

Source: Kaggle – AI-Generated Images vs Real Images

Description:

This dataset provides a balanced collection of approximately 60,000 labeled images, equally divided between AI-generated and real categories. The images cover diverse visual domains — including people, landscapes, objects, and abstract art — to ensure generalization beyond narrow or face-centric forgery detection. Each class contains an equal number of examples, supporting binary classification tasks.

Class Distribution:

The dataset was divided into 70 % train, 15 % validation, and 15 % test splits, maintaining label balance across all subsets (Figure 1).

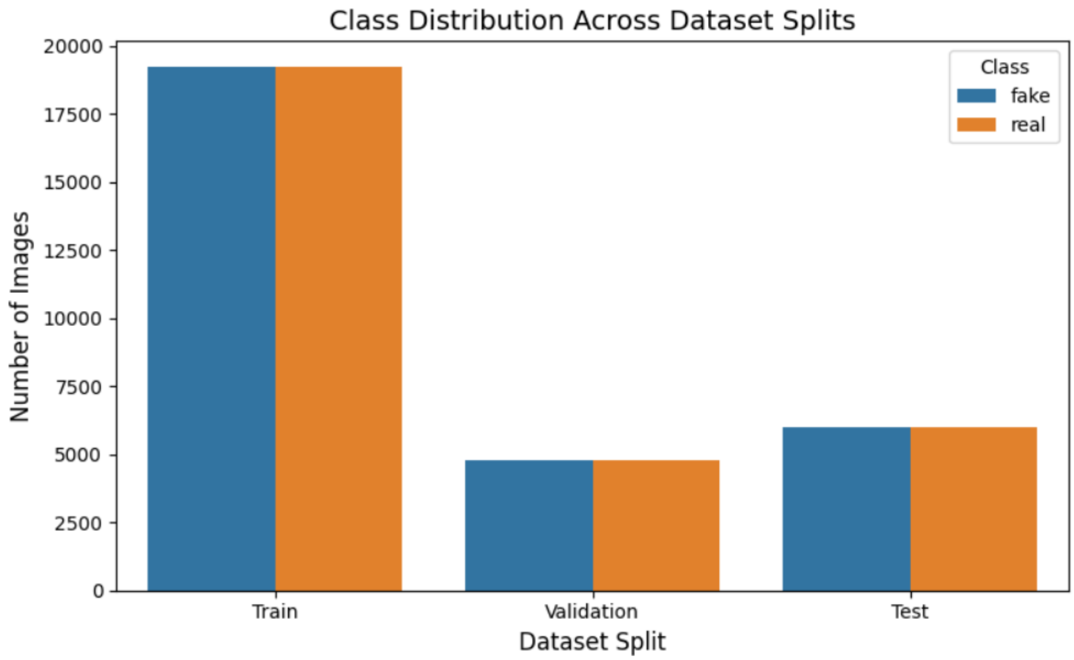


Figure 1. Class Distribution Across Dataset Splits

4.3 Text Dataset

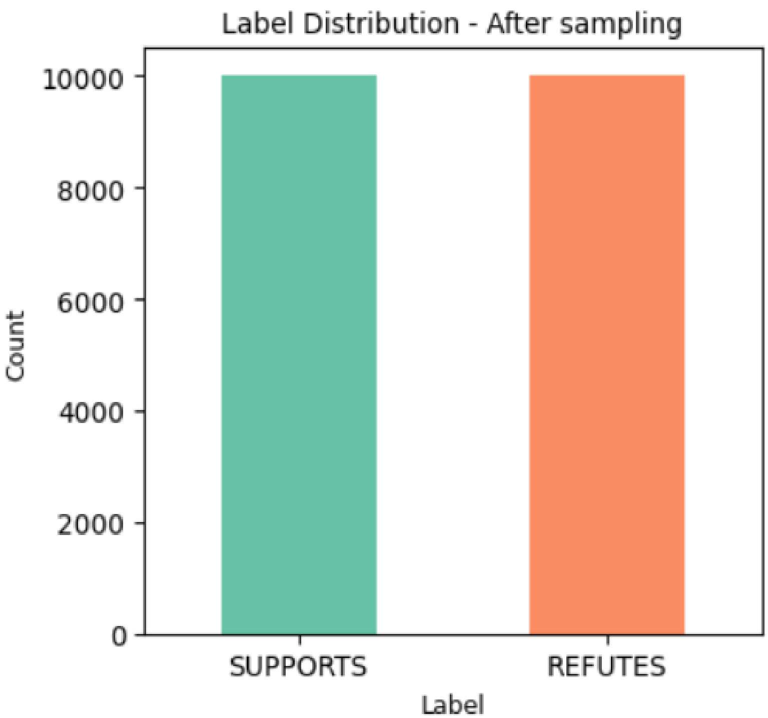
Source: Hugging Face – FEVER (Fact Extraction and VERification) Dataset

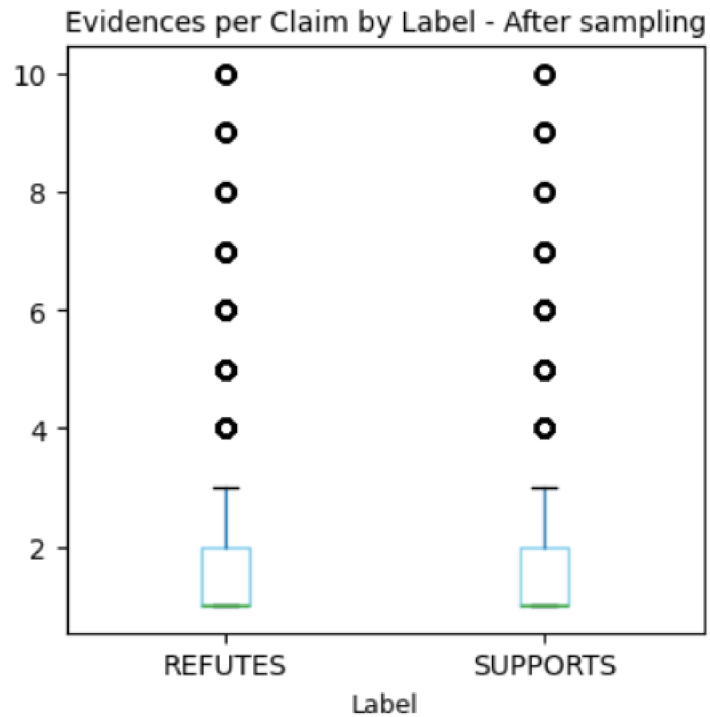
Description:

The **FEVER** dataset (Thorne et al., 2018) contains approximately **185,445 human-authored claims**, each annotated as SUPPORTED, REFUTED, or NOT ENOUGH INFO, with evidence drawn from Wikipedia. For each claim, associated evidence sentences are provided, supporting both classification and retrieval evaluation tasks.

Preprocessing and Sampling:

Initial label distribution was heavily skewed toward SUPPORTED claims. To balance classes for binary classification, NOT ENOUGH INFO examples were excluded, and down-sampling was applied to SUPPORTED examples until parity with REFUTED was achieved. Additionally, evidence sentence counts per claim were truncated to a maximum of 10 to avoid long-tail bias.





4.4 Methodology

The system implements two parallel yet complementary pipelines for misinformation detection — one for image inputs and one for text inputs. The architecture is as shown below.

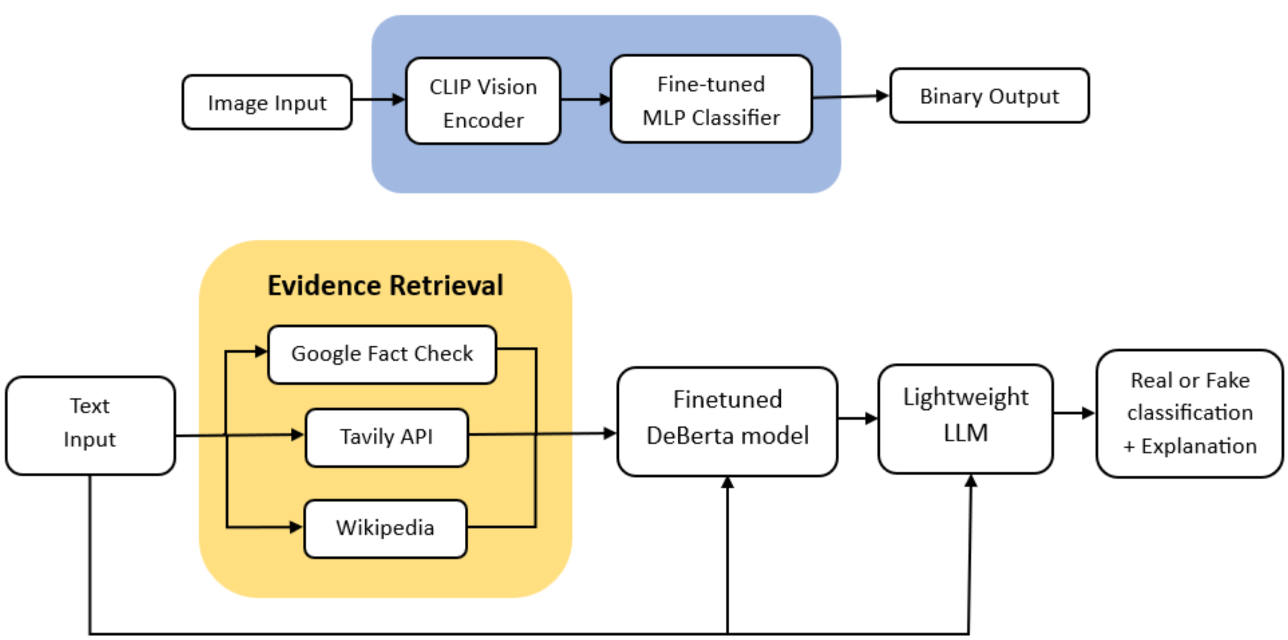


Image Pipeline:

1. Input image is preprocessed using the CLIP Vision Processor.
2. Features are extracted from the openai/clip-vit-base-patch32 encoder.
3. A fine-tuned multilayer classifier predicts real or fake with a sigmoid output.
4. The resulting probability is thresholded (0.5 → Fake, ≤ 0.5 → Real).

Text Pipeline:

1. User-entered claim is submitted for evidence retrieval using three APIs:
    i. Google Fact Check API (verified sources)
    ii. Tavily API (relevant web passages)
    iii. Wikipedia API (encyclopedic background)
2. Retrieved sentences are ranked using cosine similarity of embeddings from SentenceTransformer (**all-MiniLM-L6-v2**).
3. The top-ranked evidence and claim are concatenated and passed to a **DeBERTa-v3-base** model fine-tuned on FEVER.
4. Predicted label (Real/Fake) and evidence summary are forwarded to a lightweight LLM (**Llama 3.1 Instruct**) through the Hugging Face Inference API to generate a concise, structured explanation containing **verdict, explanation,** and **confidence**.

4.4 Summary

This multimodal approach enables complementary learning across image and text modalities. Balanced and diverse datasets ensure robust performance, while the unified pipeline architecture supports end-to-end inference through a single deployed Gradio interface. Together, the data design and methodology lay the foundation for reliable and explainable misinformation detection across visual and textual domains.

5. Model Development and Hyperparameters

5.1 Text Classifier

- **Base Model**: DeBERTa-v3-base
- **Fine-tuned Model**: rajyalakshmijampani/fever_finetuned_deberta
- **Task**: Binary classification (REAL vs. FAKE)
- **Optimizer**: AdamW (learning rate = 2e-5)
- **Epochs**: 3
- **Loss Function**: CrossEntropyLoss
- **Metrics**: Accuracy, F1 score
- **Final Accuracy**: 95%

5.2 Image Classifier

- **Base Model**: CLIP ViT-B/32
- **Custom Head**:

  Linear(1024 → 256) → ReLU → Dropout(0.5) → Linear(256 → 1) → Sigmoid Activation

- **Optimizer**: Adam (learning rate = 1e-4)
- **Epochs**: 3
- **Loss Function**: Binary Cross Entropy
- **Final Accuracy**: 96%

5.3 Explanation Model

- **Model**: meta-llama/Llama-3.1-8B-Instruct
- **API**: Hugging Face InferenceClient
- Generates structured JSON output with fields: verdict, explanation, and confidence.

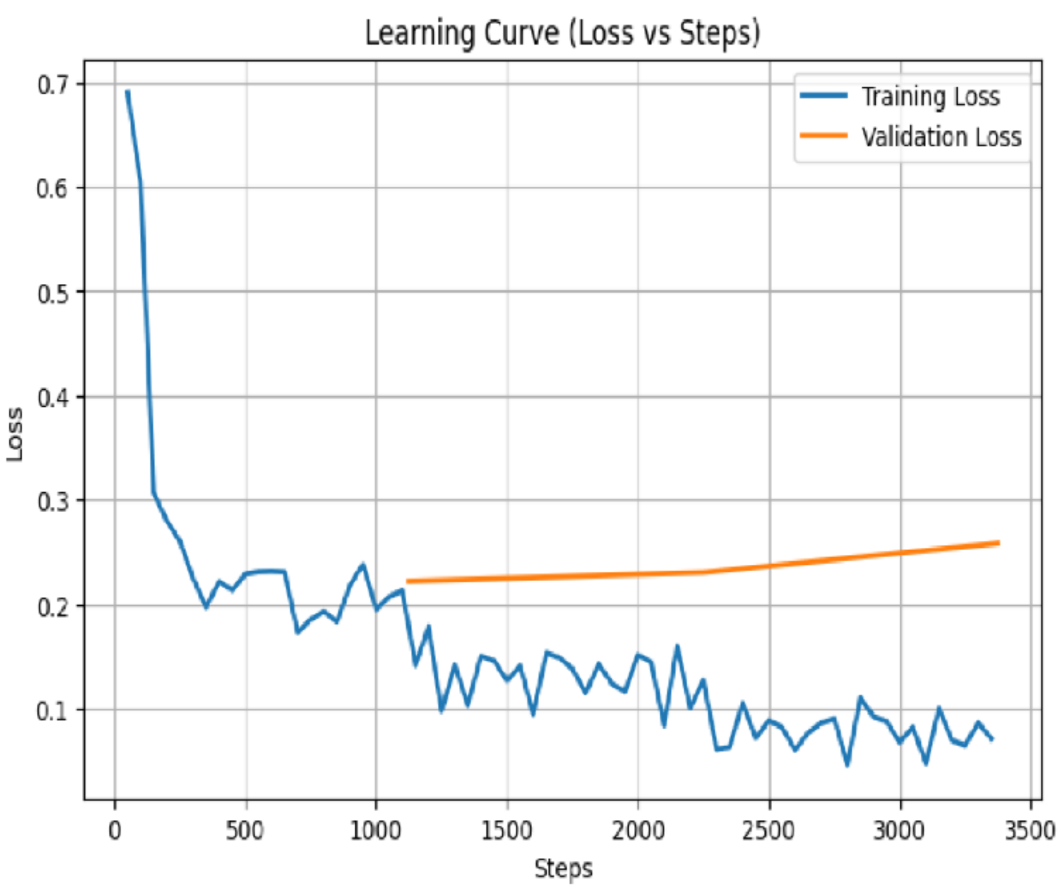
6. Evaluation and Analysis

6.1 Quantitative evaluation

- Metrics on finetuning **DeBERTa** model

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.213000	0.222147	0.948000	0.948361
2	0.127300	0.230390	0.950000	0.950199
3	0.071100	0.258387	0.950000	0.950446

```
📊 Final Evaluation Metrics:
eval_loss : 0.2584
eval_accuracy: 0.9500
eval_f1    : 0.9504
eval_runtime: 13.8990
eval_samples_per_second: 143.8950
eval_steps_per_second: 4.5330
epoch      : 3.0000
```



- Metrics on finetuning experimental **Gemma3-270M** parameter model


```
{'eval_loss': 0.303255558013916,
 'eval_accuracy': 0.8815,
 'eval_f1': 0.8802425467407782,
 'eval_precision': 0.8869653767820774,
 'eval_recall': 0.8736208625877633,
 'eval_runtime': 101.2831,
 'eval_samples_per_second': 19.747,
 'eval_steps_per_second': 1.234,
 'epoch': 1.0}
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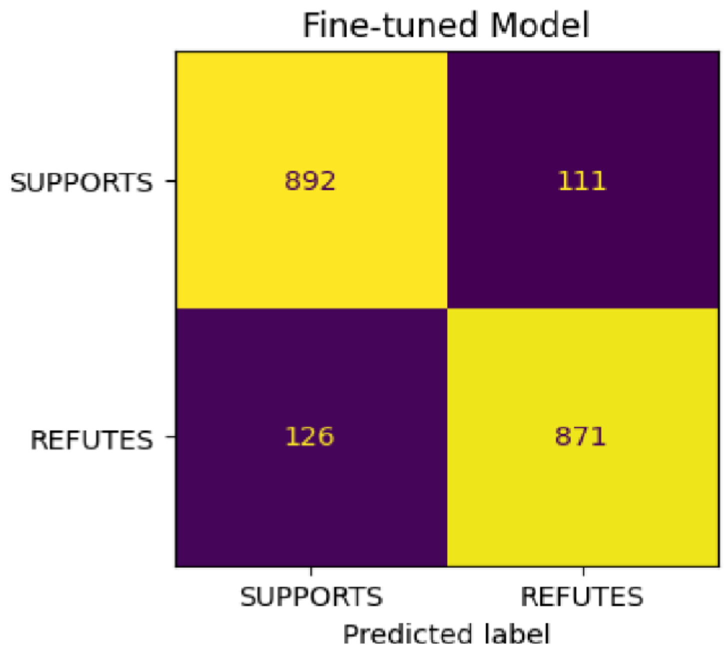
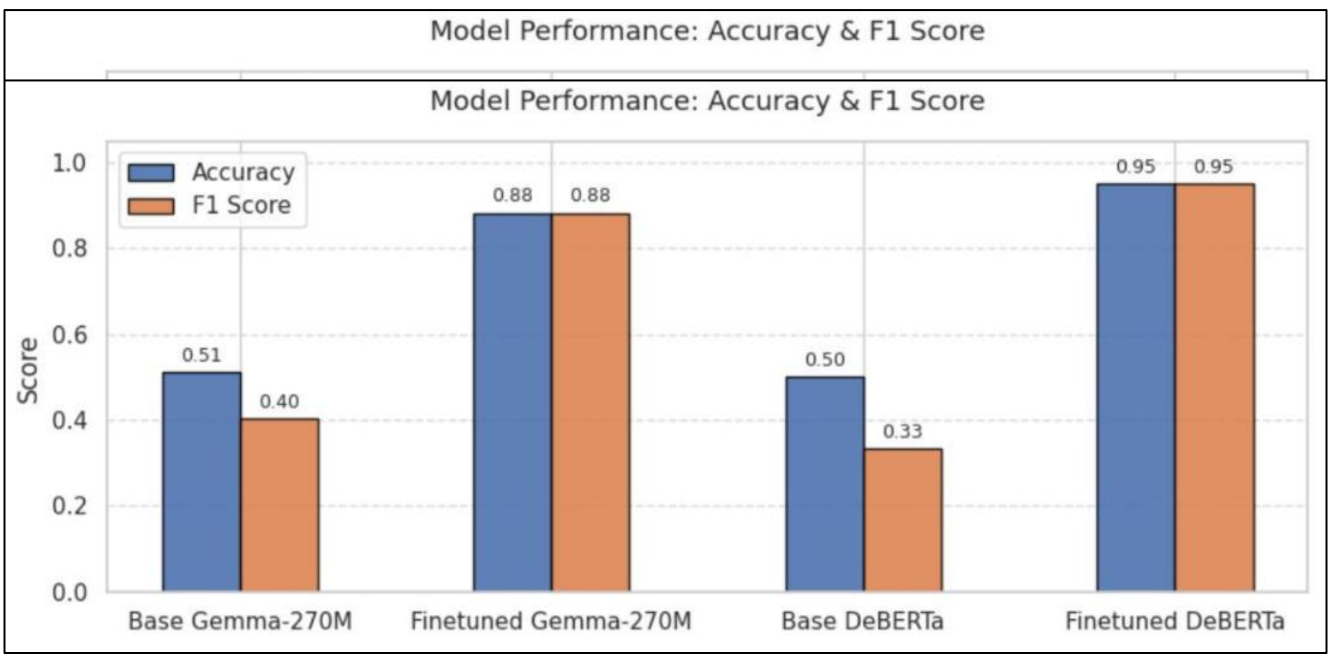


- Comparison of **DeBERTa** and **Gemma3-270M** models



- Metrics on **CLIP ViT-B/32** model

Stage	Val Loss	Val Accuracy	Train Loss	Epoch
Base (Initial)	0.6818	0.5272	–	–
Fine-tuned (Epoch 3)	0.1089	0.9591	0.1154	3

- Metrics on experimental **Convnextv-base-22k-224** model

Stage	Val Loss	Val Accuracy	Train Loss	Epoch
Base (Initial)	0.7558	0.4342	–	–
Fine-tuned (Epoch 3)	0.1834	0.9267	0.1845	3

6.2 Qualitative analysis

The text classifier accurately distinguished supported and refuted claims when evidence was strong. Failures occurred for ambiguous or humor-based claims where retrieved evidence was incomplete.

The image classifier performed well on clear manipulations but sometimes misclassified artistic filters as fakes. The LLM explanation model added interpretability, producing concise, human-readable justifications for predictions.

7. Deployment and Documentation

7.1 Deployment Platform

- **Frontend**: Gradio interface (two tabs for text and image detection).
- **Backend**: Python-based application hosted on Hugging Face Spaces.
- **Model Hosting**: All models loaded dynamically from Hugging Face Hub.
- **LLM Integration**: Uses Hugging Face Inference API for explanation generation.

7.2 Inference Pipeline

- User provides text claim or image.
- If text: evidence retrieval + DeBERTa classification + Llama explanation.
- If image: processed by CLIP vision encoder + fine-tuned classifier head.
- Results displayed in Gradio markdown output.

7.3 Example Outputs

Multimodal Misinformation Detector

Text Detector | Image Detector

Enter Claim

Sun rises in the west

Classify Claim

Prediction: The claim is Fake.

Explanation: The Sun appears to rise in the east and set in the west due to Earth's rotation. This is a widely observed phenomenon and a fundamental aspect of our planet's geography. The statement that the Sun rises in the west contradicts this established fact.

Confidence: High.

Multimodal Misinformation Detector

Text Detector | Image Detector

Enter Claim

India is in Asian continent

Classify Claim

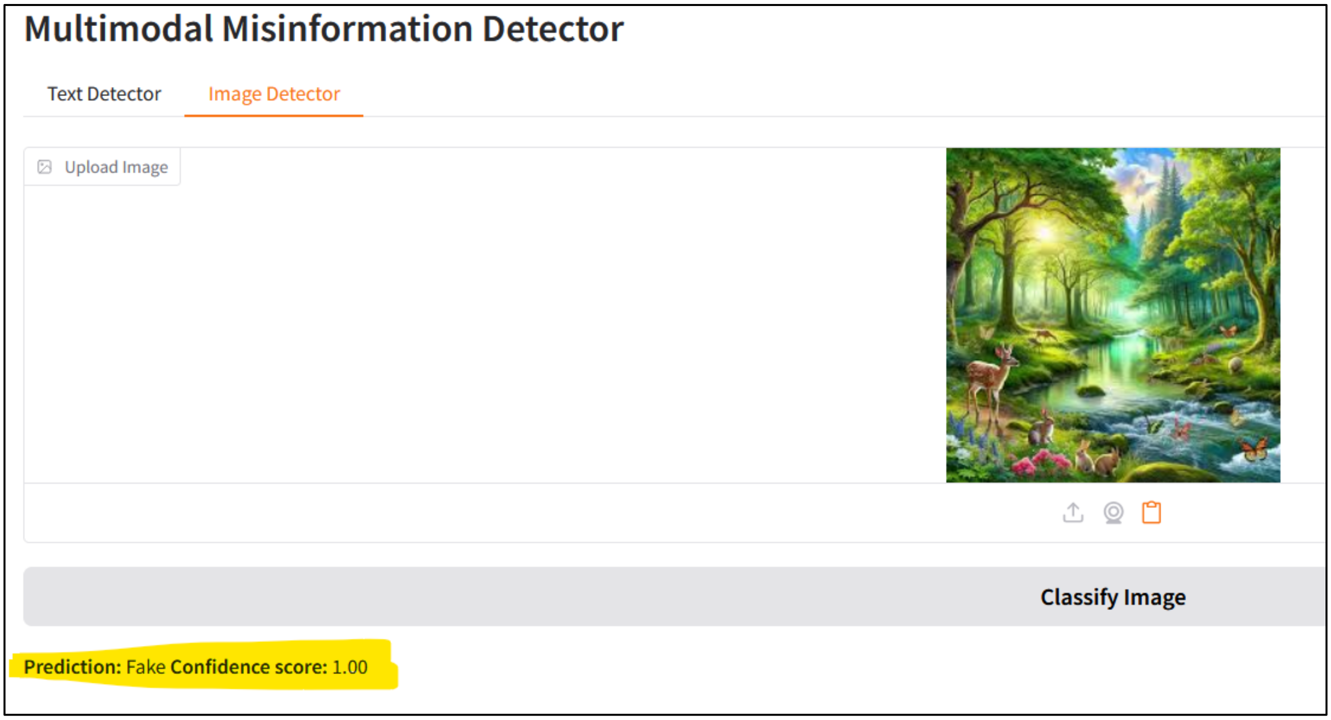
Prediction: The claim is Real.

Explanation: India is indeed located within the Asian continent, sharing borders with countries like China and Nepal. The Indian subcontinent is a distinct geographic region that is part of the larger Asian continent. This is due to its geographical features, such as the Himalayan Mountains, and its position on the Indian Tectonic Plate.

Confidence: High.

Classify Image

Prediction: Real **Confidence score:** 0.49



7.4 Documentation

Full setup, dependency, and model usage instructions are included in *README.md* and technical documentation accessible at [Github Repo]. All modules are modular and reusable for future extensions.

8. Conclusion and Future Work

This project demonstrates an end-to-end multimodal misinformation detection pipeline integrating retrieval, classification, and explainability. The combination of transformer-based NLP models and CLIP-based vision models enables more robust multimodal verification than unimodal approaches. The deployed Hugging Face Space provides a functional and user-friendly interface accessible to both technical and non-technical users.

Future Directions:

- Experiment with larger LLMs or multimodal transformers.
- Extend to video or audio misinformation detection.

- Develop REST API endpoints for programmatic access.
- Integrate user feedback to improve model accuracy over time.
- Implement real-time monitoring of usage and performance metrics.
- Set up unit and integration tests for core functions and models.

9. References

- Thorne et al., "FEVER: A Large-scale Dataset for Fact Extraction and Verification," EMNLP 2018.
- He et al., "DeBERTa: Decoding-enhanced BERT with Disentangled Attention," ICLR 2021.
- Radford et al., "Learning Transferable Visual Models from Natural Language Supervision," ICML 2021.
- OpenAI, "CLIP Model Card," 2021.
- Meta AI, "Llama 3.1 Technical Report," 2024.

10. Appendix

10.1 Training parameters

- **Text Classifier**: Learning Rate = 2e-5, Epochs = 3, Batch Size = 32.
- **Image Classifier**: Learning Rate = 1e-4, Epochs = 3, Batch Size = 32.

10.2 API References

- Google Fact Check API
- Tavily Search API
- Wikipedia API
- Hugging Face Inference API

10.3 Deployment Link

https://huggingface.co/spaces/group9-dsailab/multimodal_misinfo_detector

10.4 License Information

- **Code**: MIT License
- **Datasets**: CC BY-SA 4.0 (FEVER), CC BY-SA 3.0 (Wikipedia)
- **Models**: Licensed under respective Hugging Face terms